

opportunities here. VLC can be used for vehicular communication due to the presence of vehicle lights and existing traffic light infrastructure. High-priority applications for vehicle safety communication include collision warning, pre-crash sensing, emergency electronic brake lights, lane-change warning and curve speed warning, among others.

Such high-priority applications require reliable reachability with extremely low latency. Due to extremely low allowable latency in vehicle safety communication, a high-speed VLC system like Li-Fi can be useful.

An outdoor VLC system using controller area network (CAN) has been proposed, where backlights and headlights are used for communication.

Defence and security. The ability to send data quickly and in a secure way is the key to many applications. Visible light not being detected on the other side of a wall has great security advantages.

Hospitals and healthcare. There are advantages of using VLC in hospitals and in healthcare. Cellphones and Wi-Fi are undesirable in certain areas of hospitals, especially around MRI scanners and in operating theatres. This is where VLC can come in.

Wi-Fi spectrum relief. Wi-Fi cannot keep up with the growing demand for wireless data. VLC can provide data rates higher than Wi-Fi. This can be done at a low cost, since radio frequency (RF) components and antenna systems are eliminated.

Aviation. Radio is undesirable in passenger compartments of aircraft. LEDs are already used for illumination and can also be used instead of wires to provide media services. This reduces aircraft construction costs and weight.

Underwater communication. RF does not work underwater, but visible light can support high-speed data transmission over short distances in this environment. This could enable divers and underwater vehicles to talk to each other.

Conclusion

Although this technology is still in its infancy, with further studies and development its far-reaching applications will only get better. VLC technology is all about using LED light bulbs meant for illumination to also send data simultaneously. It is best suited as an alternative for data transfer where radio transmission networks are not desired or possible. In future, we will be able to transmit an image, audio and even a high-definition video using LED light bulbs.

Features such as high bandwidth, non-interference with radio waves (in electromagnetic-sensitive areas) and non-hazardous to health has made VLC an attractive technique for future communication. Li-Fi is 100 times faster than analogous Wi-Fi, which uses RF for communication. The optical wireless communication system is a good replacement for the regular communication system. VLC is a rapidly growing segment in the field of communication. **EFY**

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